

Cybersecurity in Supply Chain Management

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Introduction to SCM: 33:799:301:08

02/17/26

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What is Cyber Security in the Supply Chain?



- The practice of protecting systems and networks from unauthorized access or disruption.
- The goal is to preserve data integrity, confidentiality, and availability (the CIA triad).
- Strives to prevent disruptions to operations and maintain company integrity in the private and public sector

Supply Chain Cyber Security Threats

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Cyber Security Threats in the Supply Chain

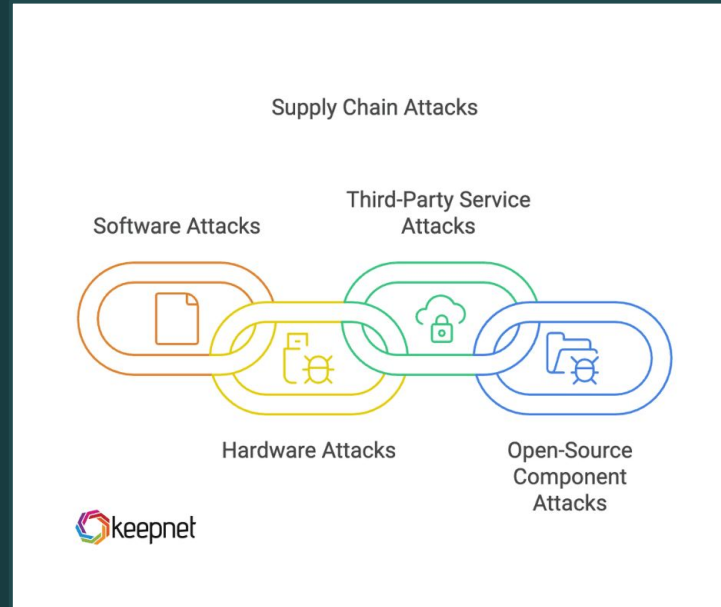
- Software Supply Chain Attacks - This occurs when there is malicious software integrated into a product or system update before it is passed on to the consumer
- Hardware Trojan/Tampering - The implementation of physical malicious hardware during the manufacturing and distribution phase of the supply chain process
- Ransomware - The emergence of a specific type of malware that encrypts data and demands payment for its release.
- Phishing and Social Engineering - This is used to manipulate holders of login credentials allow the compromising of the systems or trick them into revealing sensitive information or avenues for access.



Supply Chain Cyber Security Threats (Vulnerabilities)

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- ❖ Lack of system oversight and maintenance
- ❖ Out-of-date software within third party suppliers and manufacturers
- ❖ Consolidation of vital information with easy access permission



- ❖ Stolen login credentials of consumers and suppliers
- ❖ Outdated customer interfaces allowing for potential hacking threats
- ❖ Insufficient restrictions and oversight on product production

Recent incidents of Cyber Attacks: Jaguar Land Rover

What happened:

- August 2025 → Jaguar Land Rover suffered a cyberattack that compromised and forced a complete shutdown of their systems
 - Used social engineering to steal credentials and gain access to sensitive data
- JLR paused production at major facilities in the United Kingdom, Slovakia, India, China, and Brazil for months
 - Production lines were stopped for weeks
 - Employees couldn't go to work
 - Essential systems such as manufacture control and ordering were offline

What happened next:

- Restart of operations began in October 2025
- The company worked with cybersecurity professionals and law enforcement to investigate and restore systems before resuming operations
- JLR started a fast-pay scheme to pay suppliers earlier and ease cash-flow pressures



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As a Result:

- Jaguar Land Rover suffered a direct loss of \$258 million
- Around 27,000 vehicles were not produced
- Spent \$266.9 million in additional costs to recover
- UK car production dropped to its lowest level since 1952

Recent incidents of Cyber Attacks: Colonial Pipeline

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What happened:

- In May 2021, Colonial Pipeline was hit by a ransomware attack carried out by DarkSide, who gained access to the company's IT network through a leaked password account that didn't have multi-factor authentication.
- Affected IT systems but still chose to shut down pipeline operations as a precaution
 - Billing systems, monitoring tools, managing fuel flows or transactions



What happened next:

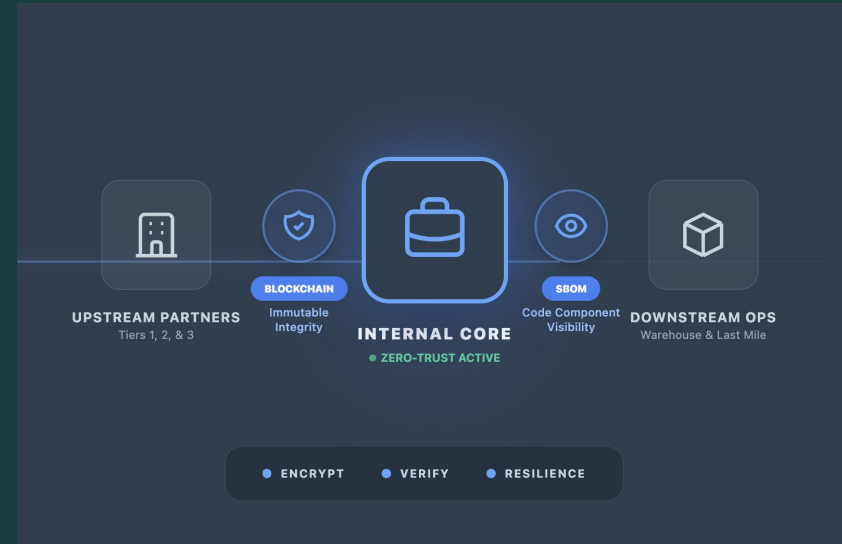
- Groups such as the FBI, CISA, and TSA were involved in investigating and recovering from this incident
- They paid \$4.4 million in Bitcoin to the attacker

As a result:

- Colonial Pipeline supplies about 45% of fuel on the East Coast so even though this shutdown lasted for 5 days, it still caused fuel shortages, panic buying, long lines at gas stations, and price increases.
- Following this incident, the US government issued new cybersecurity regulations for pipeline operators and put focus on stronger defenses

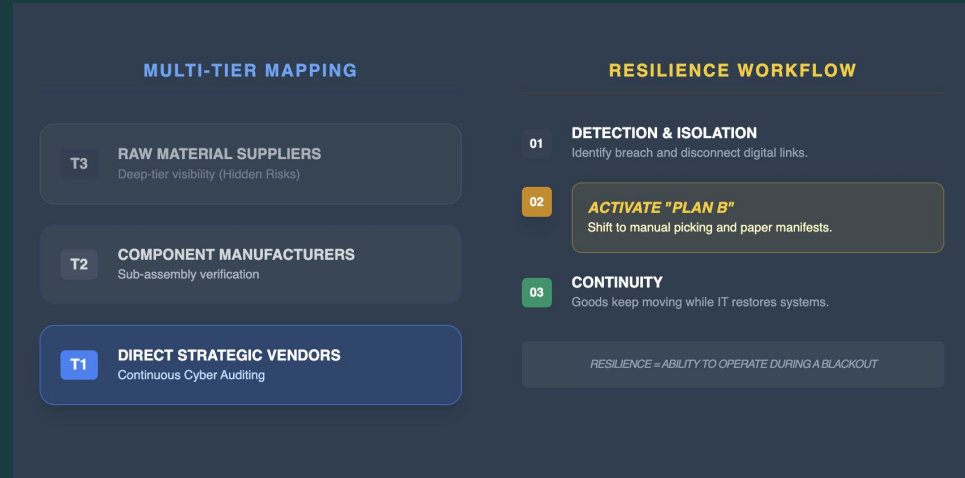
Risk Mitigation Strategies- Technical Mitigation

- **Zero Trust Architecture (ZTA):**
 - **Concept:** Shifts from a "Castle Moat" (perimeter) security model to "**Never Trust, Always Verify.**"
 - **SCM Application:** Every supplier portal and IoT device is re-authenticated for every transaction.
 - **Benefit:** Prevents "Lateral Movement"—if a Tier-3 vendor is breached, the attacker cannot jump to your central database.
- **Blockchain for Data Integrity:**
 - **Concept:** A decentralized, **immutable ledger** (records cannot be deleted or changed).
 - **SCM Application:** Tracking "Provenance" (origin) and shipping manifests in real-time.
 - **Benefit:** Prevents hackers from falsifying shipping documents or diverting payments.



Risk Mitigation Strategies- Visibility & Resilience

- **Software Bill of Materials (SBOM):**
 - **Concept:** A mandatory "ingredient list" for all software used in the supply chain.
 - **Benefit:** Allows managers to instantly identify if a global code vulnerability (like Log4j) exists within their specific warehouse systems.
- **Multi-Tier Visibility & Auditing:**
 - **Strategy:** Extending cyber-audits beyond Tier-1 (direct) partners to Tier-2 and Tier-3 sub-suppliers.
 - **Benefit:** Identifies the "Weakest Link" in the upstream supply chain before a disruption occurs.
- **Operational Resilience (The "Plan B"):**
 - **Strategy:** Developing manual/analog workarounds for automated processes.
 - **Benefit:** Ensures physical goods (trucks/ships) keep moving even if digital systems (WMS/ERP) are offline due to a cyberattack.



Companies Bolstering Supply Chain Cybersecurity

IBM

- Develops blockchain solutions to improve transparency and secure data sharing across global supply chains
- Helps track shipment provenance and prevent falsified records



Maersk

- Strengthened cybersecurity controls after the Not Petya ransomware attack
- Uses TradeLens blockchain platform to improve documentation security and shipment visibility



MAERSK

U.S. Government & Federal Agencies

- Issued Executive Order 14028 to improve cybersecurity across critical infrastructure and supply chains
- Promote Zero Trust adoption and secure software practices

Industry-Wide Adoption

- Growing use of Software Bill of Materials (SBOM) and multi-tier supplier monitoring
- Increased focus on cyber resilience and faster recovery after attacks

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Best Practices for Securing Supply Chains

Strategy & Technology

- Establish clear cybersecurity governance and define responsibilities across internal teams and suppliers
- Conduct regular vendor risk assessments to ensure suppliers meet security standards
- Increase visibility across Tier-1, Tier-2, and Tier-3 partners to identify risks earlier
- Implement layered technical defenses such as multi-factor authentication, network segmentation, and continuous monitoring
- Protect data integrity and secure information sharing across logistics, manufacturing, and procurement systems

People, Response & Resilience

- Provide regular cybersecurity training to employees and suppliers, focusing on phishing and social engineering risks
- Develop clear incident response plans to reduce downtime during cyberattacks
- Maintain backup processes and operational alternatives so supply chain activities can continue during system failures
- Shift from reactive cybersecurity toward proactive risk management and continuous improvement

Works Cited

Futterer, Alfons. "What Is a Supply Chain Attack?" *NanoMatrix Secure*, 12 Sept. 2024, www.nanomatrixsecure.com/what-is-a-supply-chain-attack-and-how-can-organizations-avoid-them/.

"Global Cybersecurity Outlook 2026." *World Economic Forum*, 12 Jan. 2026, www.weforum.org/publications/global-cybersecurity-outlook-2026/digest/.

Handfield, Robert, et al. "Https://Www.Sciencedirect.Com/Science/Article/Pii/S2211379718320771?Via=ihub | Request PDF." *Researchgate.Net*, www.researchgate.net/publication/328544309_httpswwwsciencedirectcomsciencearticlepiiS2211379718320771via3Dihub. Accessed 17 Feb. 2026.

Leggett, Theo. "Breaking News, Video and the Latest Top Stories from the U.S. and around the World." *BBC News*, BBC, www.bbc.com/news. Accessed 16 Feb. 2026.

National Institute of Standards and Technology. (2020). Zero trust architecture (NIST SP 800-207). <https://doi.org/10.6028/NIST.SP.800-207>

"Notpetya Ransomware Attack on Maersk - Key Learnings." *LRQA*, 29 Oct. 2024, www.lrqa.com/en/insights/articles/notpetya-ransomware-attack-on-maersk-key-learnings/.

Parfomak, Paul, and Chris Jaikaran. "CRS Insight Prepared for Members and Committees of Congress Insighti." *Congress.Gov*, 11 May 2021, www.congress.gov/crs_external_products/IN/PDF/IN11667/IN11667.2.pdf.

"Press Release: Gartner Identifies Top Supply Chain Technology Trends for 2025." *Gartner.Com*, 18 Mar. 2025, www.gartner.com/en/newsroom/press-releases/2025-03-18-gartner-identifies-top-supply-chain-technology-trends-for-2025.

Rose, Scott, et al. "Zero Trust Architecture." *CSRC*, 11 Aug. 2020, csrc.nist.gov/pubs/sp/800/207/final.

Sandrine. "Modernizing Global Supply Chains with Blockchain." *D'Amore-McKim School of Business*, 4 Apr. 2023, damore-mckim.northeastern.edu/news/modernizing-global-supply-chains-with-blockchain/.

"Software Bill of Materials (SBOM) | Cisa." *Cisa.Gov*, www.cisa.gov/sbom. Accessed 17 Feb. 2026.

Verma, Sanjeev. "Blockchain for Supply Chain." *IBM*, 17 Nov. 2025, www.ibm.com/think/topics/blockchain-for-supply-chain.

World Economic Forum. (2025). Global cybersecurity outlook 2026. <https://www.weforum.org/publications/global-cybersecurity-outlook-2026/>